

Hans Kersting, PhD

Research Scientist — Applied ML & Production Systems

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ML scientist with 10+ years in research and production. Track record shipping models at scale, leading cross-functional teams, and designing experiments to measure customer impact.

Skills

Programming **Python, SQL, Pytorch, Tensorflow**, Spark, Hadoop, AWS, Git
Expertise **Machine Learning, Deep Learning, Experiment Design**, Probability Theory, Bayesian Methods, Optimization
Languages Fluent **English** and **German**

Experience

- 01/2023–present **Research Scientist, Yahoo**, Berlin, Germany
ML R&D for Yahoo's demand-side platform, serving **500B+ bid requests/day** at millions of QPS.
- **Led cross-functional team** (4 engineers, 2 data scientists) to **rapidly ship** new ML product feature end-to-end—design docs to beta with **positive advertiser feedback**
 - Designed new **click prediction model** (field-weighted factorization machine), now standard across DSP: **+5.6% lower CPC, +3.5% CTR, +8% fees**
 - Co-developed novel **online SGD algorithm** for real-time bid optimization; deployed to production (runs every 5 mins)
 - Partnered with **product managers** on feasibility and timelines; designed experiments to measure customer impact
- 10/2020–12/2022 **Postdoctoral Researcher, INRIA & ENS Paris**, France
Research on **optimizers for deep learning**. Published at **ICML, NeurIPS**, and **AISTATS**. Co-organized reading groups and workshops.
- 07/2019–09/2019 **Research Intern, Amazon Research**, Cambridge, UK
Research on **ML for supply-chain optimization**.
- 10/2017–02/2018 **Research Intern, Bosch Center for AI (BCAI)**, Stuttgart, Germany
Research on **inverse problems** for engine control.

Education

- 2015–2020 **PhD in Machine Learning, Max Planck Institute & Univ. Tübingen**
Grade: **magna cum laude**. Thesis on probabilistic numerics and Bayesian inference.
- 2013–2015 **MSc in Mathematics, LMU Munich**, GPA: 1.44 (very good)
Specialization in stochastic calculus and probabilistic modelling.
- 2009–2013 **BSc in Mathematics, LMU Munich**, GPA: 1.14 (very good)
Specialization in probability theory.

Selected Publications

- Book **Probabilistic Numerics – Computation as Machine Learning**, Hennig, Osborne, **Kersting**. Cambridge University Press, 2022.
- ICML 2022 Anticorrelated Noise Injection for Improved Generalization. Orvieto*, **Kersting***, Proske, Bach, Lucchi.

- NeurIPS 2022 On the Theoretical Properties of Noise Correlation in Stochastic Optimization. Lucchi*, Proske*, Orvieto, Bach, **Kersting**.
- ICML 2020 Differentiable Likelihoods for Fast Inversion of 'Likelihood-Free' Dynamical Systems. **Kersting***, Krämer*, et al.

Full list: Google Scholar (h-index: 10, 580+ citations)

Recognition

- 2022 **Outstanding Reviewer Award, ICML**
Top 10% of reviewers.
- 2013–2015 **German Academic Scholarship Foundation, Studienstiftung**
Awarded to <0.5% of German university students.